Scalability Analysis

Current Scale

The pipeline is validated on a synthetic dataset of 300 students across 20 campuses (C01 - C20, 15 students each). At this scale, a full pipeline run completes in under 60 seconds including all 14 Anthropic API calls, and produces six output files plus nine documentation files. The entire run uses 31,771 tokens (19,693 input, 12,078 output), which at current claude-sonnet-4-5 pricing represents a negligible cost well within any reasonable operating budget.

- Students: 300 total (15 per campus × 20 campuses)

- Campuses: 20 (C01 - C20)

- LLM API calls: 14 (one per campus that has CRITICAL/HIGH students)

- Tokens consumed: 31,771 total (19,693 input + 12,078 output)

- Fallbacks triggered: 0

- Duplicate IDs removed: 9

- Pipeline runtime: under 60 seconds end-to-end

Bottlenecks at 100 Campuses

Scaling from 20 to 100 campuses (approximately 1,500 students at 15 per campus) does not change the pipeline architecture, but it does surface two bottlenecks that need to be understood before deployment at that scale: API call count and sequential I/O for the per-campus Excel dashboards.

- API calls increase linearly with active campuses. At 100 campuses with all campuses having at least one CRITICAL/HIGH student, API calls increase from 14 to approximately 70 (assuming not all campuses have at-risk students on every run). This is within Claude API rate limits for standard tiers.

- Per-campus Excel file writes are sequential. At 100 campuses, writing 100 .xlsx files adds wall-clock time. Parallelising with concurrent.futures ThreadPoolExecutor is the recommended optimisation at this scale.

- Memory footprint: the merged DataFrame at 1,500 students is approximately 2 - 3 MB in memory. pandas can handle 100,000+ rows comfortably on any modern machine; memory is not a bottleneck at 100 campuses.

- Single-file CSV input: at 100 campuses, the CSV files grow to ~22,500 rows (1,500 students × 15 days of metrics). pd.read_csv() handles this in under 1 second; no chunking needed below 1 million rows.

Migration Path

The pipeline is designed for progressive infrastructure upgrade as campus count grows. The migration path is CSV -> SQLite -> PostgreSQL; each step adds durability and query capability without requiring a rewrite of the pipeline logic, because all data access is isolated to ingestion.py.

- Phase 1 - CSV (current, 1 - 50 campuses): daily CSV exports from the student information system, ingested by ingestion.py. No server required. Single-file operation. Suitable for one part-time engineer managing up to 50 campuses.

- Phase 2 - SQLite (50 - 200 campuses): replace CSV ingestion with pd.read_sql_query() against a local SQLite database. SQLite handles 200 concurrent campus records comfortably. No server process required; the .db file travels with the project. ingestion.py is the only module that changes.

- Phase 3 - PostgreSQL (200+ campuses): replace SQLite with a managed PostgreSQL instance (e.g., Supabase free tier or Railway). Enables concurrent writes from multiple campus coordinators, row-level security for PII columns, and query-based filtering to run the pipeline for one campus subset at a time.

- No other module (risk_engine, llm_engine, output_generator, doc_generator) needs to change for any phase of this migration - they operate on the DataFrame returned by ingestion.py regardless of its source.

Cost Projection

At the current scale of 300 students and 20 campuses, the pipeline used 31,771 tokens in a single run (19,693 input + 12,078 output), giving an average of approximately 106 tokens per student (31,771 ÷ 300). Projecting to 100 campuses at 15 students each with a 40% CRITICAL/HIGH rate: 100 × 15 × 0.40 = 600 at-risk students per run. At 106 tokens per student, that is 63,600 tokens per run. Running daily for 30 days gives 1,908,000 tokens per month. At claude-sonnet-4-5 pricing (approximately $3 per million input tokens and $15 per million output tokens, blended ~$6/million for a 60/40 input/output mix), 1,908,000 tokens costs approximately $11.45 per month - well within the $200/month budget target. Even at 3x that volume (300 campuses, twice-daily runs), the cost remains under $70/month. The $200/month budget provides a 17x headroom over the projected 100-campus cost, leaving ample room for model upgrades or increased at-risk student rates.

Scale Comparison

The table below projects at-risk student count and token usage as campus count grows. Calculation basis: 106 tokens/student; 40% CRITICAL/HIGH rate assumed.

Scale	At-Risk Students/Run	Est. Tokens/Run
20 campuses (current)	~120	~12,720
100 campuses	~600	~63,600

Monthly Cost Estimate

Estimated monthly cost at claude-sonnet-4-5 pricing (~$6/million tokens blended). Both scenarios are well within the $200/month budget target - 17x headroom at 100 campuses.

Runs/Month	Tokens/Run	Est. Monthly Cost
30	~12,720	<$1/month
30	~63,600	~$5-10/month